Project Documentation

Social Media Impact on Productivity & Daily Behavior

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Track:	Data Analysis

1. Project Idea

The project revolves around analyzing the digital footprint and social media habits of university students and young professionals. By leveraging data analytics, the project visualizes how various social media platforms consume daily hours and correlates this consumption with critical lifestyle factors such as sleep quality, study/work hours, and perceived productivity. The result is a comprehensive dashboard that synthesizes survey data into clear, actionable behavioral insights.

2. Business Problem

With the exponential rise in digital content consumption, there is a growing concern regarding its negative impact on academic and professional productivity. The core problem this project solves is the lack of visibility into these habits. Users often underestimate the time they spend on platforms like TikTok and Instagram. This project provides a clear, data-driven reflection of how social media usage drains time, decreases sleep hours (often to less than 5 hours), and hinders daily goals.

3. Tools & Technologies Used

- **Microsoft Power BI:** Used for data modeling, DAX measure creation, and designing the interactive dashboard.
- **Power Query:** Utilized for data cleaning, transforming raw survey responses, and handling missing or inconsistent data.
- **Microsoft Excel / CSV:** Used as the primary database storage for the raw datasets (DEPI_Prj_FullData, Content_Type, Usage_Reasons).
- **Canva / Design Tools:** Used for creating the custom color palette, background templates, and UI/UX elements of the dashboard.

4. Project Components

A. Main Page (Overview)

The landing page establishes the demographic baseline. It visualizes the total number of responses (437), gender distribution (62.7% Female), age brackets (dominantly 18-22), current status (mostly Students), and the most used platforms (led by TikTok).

B. System Pages (Analytical Views)

- **Usage Patterns:** Analyzes behavioral habits, showing peak usage hours (mostly during free time) and preferred content types.
- **The Impact:** The core analytical page correlating social media hours with daily sleep and study/work hours to highlight productivity loss.
- **Deep Dive:** Employs advanced scatter plots and matrices to cross-filter variables for in-depth insights.

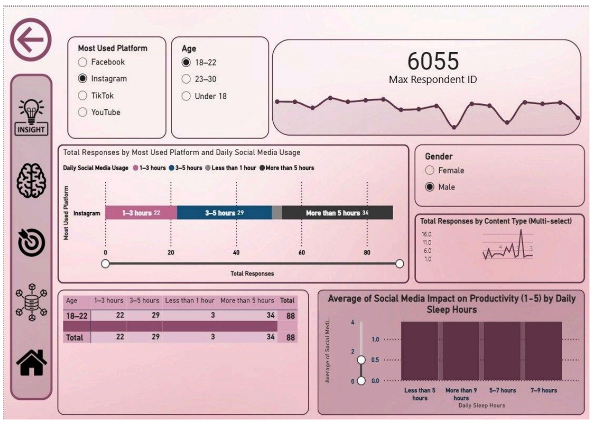
C. Core Functions

The dashboard relies on interactive slicers allowing users to filter by Gender, Age, and Platform across all pages. Complex DAX measures were created to calculate percentages, rank content quality ratings, and group hours into logical buckets.

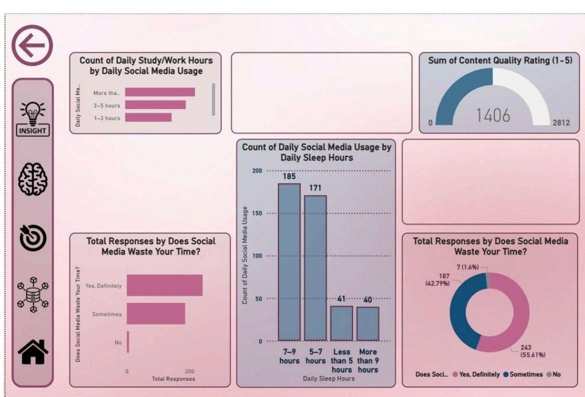
D. Database Structure

The project utilizes a Star Schema model centered around the main fact table (DEPI_Prj_FullData.csv), connected to dimensional tables like Content_Type and Usage_Reasons to categorize responses efficiently.

5. Screenshots & Screen Functions

<table>
<tr><th>Screen / View</th><th>Function & Brief Description</th></tr>
<tr><td>


</td><td>Overview: Displays high-level KPIs, demographic breakdowns (Age, Gender, Status), and ranks top platforms.</td></tr>
<tr><td>


</td><td>Usage Patterns: Highlights when users are most active and what content they consume.</td></tr>
<tr><td>


</td><td>The Impact: Correlates digital habits with physical well-being, specifically showing how usage affects sleep hours and perceived time-wasting.</td></tr>
</table>

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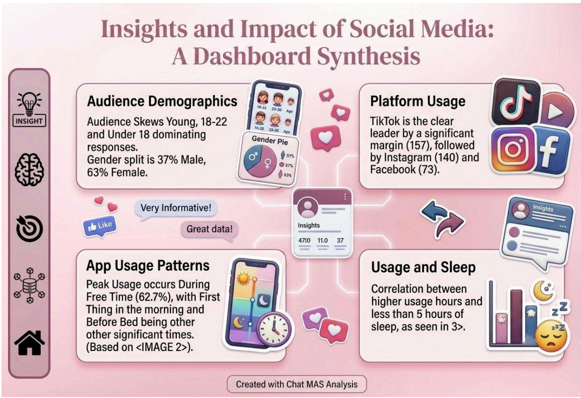
<tr><th>Screen / View</th><th>Function & Brief Description</th></tr>
<tr><td>
</td><td>Insights Synthesis: A summary graphic created to easily communicate the top 4 findings to stakeholders (Demographics, App Usage, Platform Leaders, Usage vs. Sleep).</td></tr>
</table>

6. Challenges & Solutions

- **Challenge: Messy Survey Data.** Raw data from the survey contained inconsistent text entries and unstandardized formats.
  Solution: Utilized Power Query for robust data cleaning, replacing values, and standardizing text to ensure accurate aggregations.
- **Challenge: Creating Meaningful Correlations.** It was difficult to visually show the relationship between sleep deprivation and social media usage.
  Solution: Designed a specific clustered bar chart that plotted daily sleep hours against daily social media usage, visually proving the correlation.
- **Challenge: UI/UX Consistency.** Standard Power BI visuals can look rigid.
  Solution: Implemented a custom color palette (maroon/pink theme) and imported background templates designed externally to give the dashboard a modern, cohesive identity.

7. Future Enhancements

Moving forward, the project could be expanded by:

- Integrating real-time data via APIs to monitor live usage trends instead of relying on static survey responses.
- Applying Machine Learning (via Python integration in Power BI) to predict productivity drops based on specific usage patterns.
- Expanding the survey demographic to include different age groups (e.g., 30-40 years old) to compare generational digital habits.

8. Conclusion

The "Impact of Social Media" dashboard successfully translates raw behavioral data into a compelling narrative. The results confirm a stark reality: 55.6% of respondents definitively feel social media wastes their time. Platforms like TikTok heavily dominate user attention, predominantly during free time, leading to a noticeable reduction in sleep (often under 5 hours) and productivity. This analytical tool not only highlights the problem but provides a data-driven foundation for promoting better digital wellbeing and time management among youth.